Davi Ferreira

Software Engineering Student | UFAM - IComp - Digital Processes Intern - Michelin Manaus

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PROFESSIONAL SUMMARY

Software Engineering undergraduate at the Federal University of Amazonas (UFAM) and Digital Processes Intern at Michelin. Proven experience developing shop-floor industrial applications (React, Power Apps, Dataverse), gathering operator requirements, automating enterprise workflows, and structuring relational datasets for manufacturing reliability and end-to-end traceability.

PROFESSIONAL EXPERIENCE

Michelin - Digital Processes Intern (January 2026 - Present | Manaus, Brazil)

- **Industrial Press Downtime Application: Built with React and Microsoft Dataverse across 48 curing presses.**
  Conducted user research and usability testing with operators, eliminating manual paper and spreadsheet logs.
- **Tire Quality Inspection Application: Built with Power Apps for scrap reporting and defect inspection.**
  Engineered logical integration linking mold identifiers directly to press production records for traceability.
- **Built event-driven automations with Power Automate and managed SharePoint enterprise access governance.**

EDUCATION

Federal University of Amazonas (UFAM) - Institute of Computing (IComp)

B.S. in Software Engineering (2024 - In progress | Manaus, Brazil)

- **Software Engineering & Dev: Fundamentals of SE, SE I & II, OOP, Software Verification & Validation, Project Management.**
- **Systems & Infrastructure: Operating Systems I, Computer Networks, Intro to Computer Organization, Advanced Programming Lab.**
- **Data & Algorithms: Algorithms & Data Structures I & II, Intro to Databases, Data Science, Programming Languages, Statistics.**
- **Ongoing (Term 2026.2): Design & Analysis of Algorithms, Scientific Methodology, Cybersecurity, Software Processes, Config Manageme**

KEY PROJECTS

- **Multithreaded TCP Socket Battleship in C (C, POSIX pthreads, mutex locks, Makefile, Linux):**
  Concurrent client-server architecture with custom text protocol arbitrating matches between 2 clients (UFAM 2025.1).
- **ENEM 2019 National Exam Data Analysis (Python, Pandas, NumPy, Matplotlib, Jupyter Notebook):**
  Exploratory data analysis correlating socioeconomic indicators with academic scores on large datasets.
- **Movie Streaming Relational Database (MySQL, 3NF Relational Modeling, DDL/DML, Analytical Queries).**
- **Minesweeper in C (Dynamic Heap Memory Allocation, Recursive Flood-Fill Algorithm, Pointer Matrices).**
- **Law Practice Web Landing Page (React, Vite - Active development with AI assistance).**

TECHNICAL SKILLS

- **Languages: C/C++, Python, JavaScript, SQL, Java.**
- **Web Engineering: React, HTML5, CSS3, Vite, Next.js, Tailwind CSS.**
- **Databases & Data: MySQL, Relational Modeling (ERD), Microsoft Dataverse, Pandas, NumPy, Matplotlib.**
- **Enterprise Platforms: Power Apps, Power Automate, SharePoint, Git, GitHub, Linux, Jupyter Notebook.**
- **Core CS: Concurrent Programming, TCP Sockets, Threads/Mutex, Data Structures, Algorithms.**